

20. Perform Sharpening of Image using Laplacian mask with negative center coefficient.

Aim:

To perform image sharpening using a Laplacian mask with a negative center coefficient using the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

```
import cv2
```

```
import numpy as np
```

```
cap=cv2.VideoCapture(0)
```

```
while True:
```

```
    ret,frame=cap.read()
```

```
    if not ret:
```

```
        break
```

```
rows,cols=frame.shape[:2]
```

```
pts1=np.float32([[50,50],[cols-50,50],[50,rows-50],[cols-  
50,rows-50]])
```

```
pts2=np.float32([[0,100],[cols,0],[100,rows],[cols-100,rows]])
```

```
M=cv2.getPerspectiveTransform(pts1,pts2)
```

```
dst=cv2.warpPerspective(frame,M,(cols,rows))
```

```
cv2.imshow("Original",frame)
```

```
cv2.imshow("Perspective Video",dst)
```

```
if cv2.waitKey(1)&0xFF==27:
```

```
    break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

Sample Input



Sample Output



Result:

The input image was successfully **sharpened using a Laplacian mask** with a negative center coefficient.